Karthishvar Aiyyar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.57
DPS Nacharam, Hyderabad	Class XII (CBSE)	2022	88.00%
DPS Nacharam, Hyderabad	Class X (CBSE)	2020	88.00%

EXPERIENCE

Site Operations & Digital Systems, Family Construction Company

- Role: Work within the family's construction company, learning site operations first-hand: foundation work, reinforcement and concreting practice, and how supervisors, labour and site cash flow are managed across sites.
- Impact: Digitised the supervisors' WhatsApp-based cash book by building Site Bill Tracker, an offline-first PWA for logging income, expenditure and bill photos, with admin verification and monthly settlements (see Projects).
- Expertise gained: Studied deep foundation engineering (IS 2911: group efficiency, reinforcement detailing, cap design, bentonite-slurry and tremie concreting) and geotechnical fundamentals against site observations.

SKILLS

Technical Skills:

- **Languages:** Python, SQL, JavaScript, TypeScript, C++
- **Web & Frontend:** React, Next.js, Tailwind CSS, Framer Motion, Vite, responsive UI, PWAs
- **Backend & Databases:** Node.js, Supabase (PostgreSQL), Firebase (Auth, Firestore, Storage)
- **Offline-First & Mobile:** Service Workers, IndexedDB, localStorage caching, Capacitor (Android)
- **Computer Vision & Algorithms:** MediaPipe Face Landmarker, CIELAB/Delta E, Levenshtein fuzzy search, rule-based recommendations
- **Automation & Documents:** OOXML/DOCX automation, LibreOffice, SVG generation, Node.js/Python pipelines
- **Construction & Operations:** Site operations, foundation engineering (IS 2911), contractor compliance, GST, TDS §194C, BOCW Act, EPF/ESI
- **Tools:** Git, GitHub, Vercel, AI-assisted development
- **Spoken:** Tamil (Native), English, Telugu, Hindi

Leadership & Soft Skills:

- **Leadership & Ownership:** Independently developed three products end-to-end (Vesto, Vesto Admin, Site Bill Tracker); served as Class Representative and Hostel Representative; coordinated workshops at Elan & nVision.
- **Client & Stakeholder Communication:** Delivered a 306-page technical manual incorporating client feedback; translated supervisors' cash-book workflow into a functional application.
- **Teamwork & Collaboration:** Three consecutive years on the Inter IIT football team; coordinated speakers, participants and logistics for campus workshops.
- **Quality & Adaptability:** Resolved document and UI issues through structured review cycles; independently developed expertise in computer vision, offline-first systems and construction compliance.

PROJECTS

Vesto: Skin-Tone-Based Fashion Platform

- Built an in-browser skin-tone detector using MediaPipe Face Landmarker, HSV brightness normalisation, CIELAB and Delta E to match facial samples to six Fitzpatrick tones; supports live camera and photo uploads.
- Developed a rule-based styling engine powering Complete Your Look, Daily Outfit and Occasion Planner for 7 occasions, with a weighted Style Score and wardrobe gap alerts.
- Built a Firestore-backed virtual wardrobe with transactional updates, optimistic caching and guest-to-account merge; added typo-tolerant search, filters, sorting and affiliate Buy links.
- Shipped as a mobile-first PWA with passwordless/Google authentication; ~9,800 lines of TypeScript across 55+ commits, with Lighthouse-driven performance improvements.
- Tech: Next.js 16, React 19, TypeScript, Tailwind CSS, Framer Motion, Firebase, MediaPipe, Vercel

Site Bill Tracker: Offline-First Cash-Book PWA

- Built for a pile-foundation construction firm to replace WhatsApp-based reporting, enabling supervisors to record income, expenditure and bill photos, with admin verification and monthly settlements.
- Engineered an offline-first architecture using IndexedDB sync queues, localStorage caching and service workers for reliable operation on low-connectivity sites.
- Developed a 25+ endpoint Node/Express REST API on Supabase with PostgreSQL/Storage, JWT role-based authentication, bcrypt PIN hashing, rate limiting and secure bill uploads.
- Added cross-supervisor labour-payment lookup and packaged the PWA for Android using Capacitor.
- Tech: React 19, Vite, Node.js, Express, Supabase, PostgreSQL, IndexedDB, Capacitor, JWT, bcrypt

POSITIONS OF RESPONSIBILITY

Class Representative, IIT Hyderabad

- Acted as the communication link between students and faculty, coordinating academic updates, schedules and announcements
- Collected class feedback, helped resolve issues, and supported the organisation of department events and meetings

Hostel Representative, IIT Hyderabad

- Point of contact between hostel residents and administration for maintenance requests, safety concerns and day-to-day issues
- Helped organise hostel activities and resolve resident complaints

EXTRACURRICULAR

Football, Inter IIT Sports Meet: Represented IIT Hyderabad on the Inter IIT football team for three consecutive years, balancing training and inter-institute tournaments with academics.